



Molax[®]-M

PRODUCT NAME

MOLAX-M FILM COATED TABLET

NAME AND STRENGTH OF ACTIVE SUBSTANCE

Each film-coated tablet contains 10 mg Domperidone Maleate

PRODUCT DESCRIPTION

White, round and biconvex film coated tablet with "m" on one side and SIAM logo on the other

PHARMACODYNAMICS

Domperidone is a dopamine antagonist with anti-emetic properties. It does not readily cross the blood-brain barrier. In domperidone users, especially in adults, extrapyramidal side effects are very rare, but domperidone promotes the release of prolactin from the pituitary. Its anti-emetic effect may be due to a combination of peripheral (gastrokinetic) effects and antagonism of dopamine receptors in the chemoreceptor trigger zone, which lies outside the blood-brain barrier in the area postrema. Animal studies, together with the low concentrations found in the brain, indicate a predominantly peripheral effect of domperidone on dopamine receptors. Studies in human have shown oral domperidone to increase lower oesophageal pressure, improve antroduodenal motility and accelerate gastric emptying. There is no effect on gastric secretion.

PHARMACOKINETICS

Absorption

Domperidone is rapidly absorbed after oral administration with peak plasma concentrations at approximately 1 hr after dosing. The C_{max} and AUC values of domperidone increased proportionally with dose in the 10 mg to 20 mg dose range. A 2- to 3-fold accumulation of domperidone AUC was observed with repeated four times daily (every 5 hr) dosing of domperidone for 4 days. The low absolute bioavailability of oral domperidone (approximately 15%) is due to an extensive first-pass metabolism in the gut wall and liver.

Although domperidone's bioavailability is enhanced in normal subjects when taken after a meal, patients with gastrointestinal complaints should take domperidone 15-30 minutes before a meal. Reduced gastric acidity impairs the absorption of domperidone. Oral bioavailability is decreased by prior concomitant administration of cimetidine and sodium bicarbonate. The time of peak absorption is slightly delayed and the AUC somewhat increased when the oral drug is taken after a meal.

Distribution

Domperidone is 91% to 93% bound to plasma proteins. Domperidone does not readily cross the blood-brain barrier. In humans, the apparent volume of distribution is 440 L. Small amounts of domperidone are distributed into breast milk; concentrations are 10% to 50% of those in maternal serum.

Metabolism

It undergoes rapid and extensive hepatic metabolism. The main metabolic pathways are N-dealkylation by cytochrome P450 isoenzyme CYP3A4, and aromatic hydroxylation by CYP3A4, CYP1A2, and CYP2E1.

Excretion

Urinary and faecal excretions amount to 31% and 66% of the oral dose respectively. The proportion of the drug excreted unchanged is small (10% of faecal excretion and approximately 1% of urinary excretion). Domperidone has an elimination half-life of about 7 to 9 hours but is prolonged in patients with severe renal insufficiency.

INDICATION

It is indicated for the relief of the symptoms of nausea and vomiting.

This includes:

- nausea and vomiting of functional, organic, infectious, or dietetic origin.
- nausea and vomiting induced by radiotherapy or drug therapy, and dopamine agonist (such as L-dopa and bromocriptine) used in the treatment of Parkinson's disease.

DOSAGE AND ADMINISTRATION

It is recommended to take Molax-M 1 tablet (10 mg) 3-4 times daily 15-30 minutes before meals. If taken after meals, absorption of the drug is somewhat delayed.

Adults and adolescents ≥ 12 years and weighing ≥ 35 kg and children weighing ≥ 35 kg

The dose of Molax-M should be lowest effective dose for the individual situation (typically 30 mg/day) and can be increased if necessary to a maximum daily oral dose of 40 mg. Usually, the maximum treatment duration should not exceed one week for treatment of acute nausea and vomiting. If nausea and vomiting persist for longer than one week, patients should consult their physician. For other indications, the initial duration of treatment is up to 4 weeks. If treatment exceeds 4 weeks, patients should be re-evaluated and the need for continued treatment reassessed.

Formulation (domperidone per unit)	Dosage	Maximum dose per day
Film-coated tablets (10 mg/tablet)	1 tablet 3-4 times per day	40 mg (4 x 10 mg tablet)

Renal impairment

Since the elimination half-life of domperidone is prolonged in severe renal impairment (serum creatinine > 6 mg/100 mL, i.e. > 0.6 mmol/L), the dosing frequency of Molax-M should be reduced to once or twice daily, depending on the severity of the impairment, and the dose may need to be reduced. Patients with severe renal impairment should be reviewed regularly.

Hepatic impairment

Molax-M is contraindicated for patients with moderate (Child-Pugh 7 to 9) or severe (Child-Pugh > 9) hepatic impairment. Dose adjustment is not required for patients with mild (Child-Pugh 5 to 6) hepatic impairment.

CONTRAINDICATIONS

Molax-M is contraindicated in the following situations:

- Known hypersensitivity to domperidone or any of the excipients
- Prolactin-releasing pituitary tumour (prolactinoma)
- In patients who have known existing prolongation of cardiac conduction intervals, particularly QTc, patients with significant electrolyte disturbances or underlying cardiac diseases such as congestive heart failure (see *Warnings and Precautions*)
- Co-administration with QT-prolonging drugs
- Co-administration with potent CYP3A4 inhibitors regardless of their QT-prolonging effects
- Whenever stimulation of gastric motility might be dangerous, e.g., in the presence of gastro-intestinal haemorrhage, mechanical obstruction or perforation
- In patients with moderate or severe hepatic impairment

Actual Size 100 %

DRAWING NO. MOL-M-10-02-L-ENG-MY-A0007

SENT DATE 19/04/2022

WARNINGS AND PRECAUTIONS

Cardiovascular effects

Domperidone has been associated with prolongation of the QT interval on the electrocardiogram. During post-marketing surveillance, there have been very rare cases of QT-prolongation and torsade de pointes in patients taking domperidone.

These reports included patients with confounding risk factors, electrolyte abnormalities and concomitant treatment which may have been contributing factors (see *Undesirable Effects*). Epidemiological studies showed that domperidone was associated with an increased risk of serious ventricular arrhythmias or sudden cardiac death (see *Undesirable Effects*). A higher risk was observed in patients older than 60 years, patients taking daily doses greater than 30 mg, and patients concurrently taking QT-prolonging drugs or CYP3A4 inhibitors. Domperidone should be used at the lowest effective dose in adults and children.

Domperidone is contraindicated in patients with known existing prolongation of cardiac conduction intervals, particularly QTc, in patients with significant electrolyte disturbances (hypokalaemia, hyperkalaemia, hypomagnesaemia), or bradycardia, or in patients with underlying cardiac diseases such as congestive heart failure due to increased risk of ventricular arrhythmia (see *Contraindications*).

Electrolyte disturbances (hypokalaemia, hyperkalaemia, hypomagnesaemia) or bradycardia are known to be conditions increasing the proarrhythmic risk.

Treatment with domperidone should be stopped if signs or symptoms occur that may be associated with cardiac arrhythmia, and the patients should consult their physician. Patients should be advised to promptly report any cardiac symptoms.

INTERACTIONS WITH OTHER MEDICATIONS

The main metabolic pathway of domperidone is through CYP3A4. In vitro and human data show that the concomitant use of drugs that significantly inhibit this enzyme may result in increased plasma levels of domperidone. Co-administration of domperidone with potent CYP3A4 inhibitors which have been shown to cause QT interval prolongation is contraindicated (see *Contraindications*).

Caution should be exercised when domperidone is co-administered with drugs which have been shown to cause QT interval prolongation and patients should be monitored closely for signs or symptoms of cardiovascular adverse reactions (see *Undesirable Effects*). Examples include:

- anti-arrhythmic class IA (e.g. disopyramide, quinidine)
- anti-arrhythmic class III (e.g. amiodarone, dofetilide, dronedarone, ibutilide, sotalol)
- certain antipsychotics (e.g. haloperidol, pimozide, sertindole)
- certain antidepressants (e.g. citalopram, escitalopram)
- certain antibiotics (e.g. levofloxacin, moxifloxacin)
- certain antifungal agents (e.g. pentamidine)
- certain antimalarial agents (e.g. halofantrine)
- certain gastro-intestinal drugs (e.g. dolasetron)
- certain drugs used in cancer (e.g. toremifene, vandetanib)
- certain other drugs (e.g. bepridil, methadone)

PREGNANCY AND LACTATION

Pregnancy

There are limited post-marketing data on the use of domperidone in pregnant women. The potential risk for humans is unknown. Therefore, MOLAX-M should only be used during pregnancy when justified by the anticipated therapeutic benefit.

Lactation

The amount of domperidone that could be ingested by an infant through breast milk is low. The maximal relative infant dose (%) is estimated to be about 0.1% of the maternal weight-adjusted dosage. It is not known whether this is harmful to the newborn. Therefore, breast-feeding is not recommended for mothers who are taking MOLAX-M.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Dizziness and somnolence have been observed following use of domperidone. Therefore, patients should be advised not to drive or use machinery or engage in other activities requiring mental alertness and coordination until they have established how MOLAX-M affects them.

UNDESIRABLE EFFECTS

The following adverse reactions have been reported in patients receiving loperamide and are classified according to system organ.

System Organ	Adverse Reactions
Nervous system disorders	: Headache, somnolence
Psychiatric disorders	: Loss of libido, anxiety, depression
Gastrointestinal disorders	: Diarrhoea
Skin and subcutaneous tissue disorders	: Rash, pruritus, urticaria
Reproductive system and breast disorders	: Galactorrhoea, breast pain, breast tenderness, gynaecomastia, menstruation irregular
General disorders	: Asthenia
Cardiac disorders	<u>Post-marketing experience (very rare)</u> : Ventricular arrhythmias, QT/QTc prolongation, Torsade de Pointes, sudden cardiac death (see <i>Warnings and Precautions</i>)

OVERDOSE AND TREATMENT

Overdose

Overdose has been reported primarily in infants and children. Symptoms of overdose may include agitation, altered consciousness, convulsion, disorientation, somnolence and extrapyramidal reactions.

Treatment

There is no specific antidote to domperidone. Close medical supervision and supportive therapy is recommended. Anticholinergic or anti-Parkinson drugs may be helpful in controlling the extrapyramidal reactions. It is advisable to contact a poison control center to obtain the latest recommendations for the management of an overdose.

STORAGE CONDITIONS

Store below 30°C

SHELF LIFE

4 years

DOSAGE FORMS AND PACKAGING AVAILABLE

Blister strips with 10 x 10 mg tablets (One box contains 100 film-coated tablets)

DATE OF REVISION OF PI

Date of revision: 25/03/2022

Manufacturer:- SIAM BHEASACH CO., LTD.

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Product Registration Holder, Importer:- SPG PHARMA (MALAYSIA) SDN. BHD.

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